

## Applications of fractions



- 1 Sharon walks  $\frac{2}{5}$  of a kilometre and runs  $\frac{1}{7}$  of a kilometre to get to school. How many kilometres does she travel?
- 2 Carl has  $\frac{3}{7}$  m of ribbon. After he uses some ribbon for a present, he has  $\frac{1}{4}$  m left.  
How much ribbon did he use on the present?
- 3 For a project, Ivan needs to cut a metal rod such that the longer piece is three times the length of the shorter piece.
  - a What fraction of the rod's total length will be made up of the shorter piece?
  - b If the rod is  $\frac{3}{8}$  m long, what is the length of the shorter piece?
- 4 Bianca has  $\frac{5}{6}$  kg of beads. She shares the beads evenly between 3 friends. How many kilograms does each friend receive?
- 5 At a party, Bill makes a drink by combining  $5\frac{1}{3}$  L of water with  $1\frac{1}{2}$  L of cordial. What is the total amount of litres in the mixed drink?
- 6 Skye wants to decorate her window sill with fairy lights. The window sill is  $3\frac{4}{9}$  m wide. If she only has  $2\frac{1}{8}$  m of fairy lights, how much of the window sill will not be decorated?
- 7 Katrina takes  $3\frac{2}{3}$  minutes to drive from her home to the local shopping centre. She spends  $\frac{1}{4}$  of this time waiting at traffic lights. Find the number of minutes she spends waiting.
- 8 Jack is making bags for his friends. He has  $3\frac{1}{2}$  m of fabric. If each bag requires  $\frac{2}{5}$  m of fabric, how many bags can he make?
- 9 At the Easter Show, Tobias spends  $\frac{2}{4}$  of an hour visiting chickens and  $\frac{3}{9}$  of an hour visiting alpacas. How much time has he spent visiting animals?
- 10 Caitlin runs a cafe and is interested in the drink choices of her customers. She finds that 3 out of 4 customers order hot chocolate. If she has 4000 customers on a particular day, how many hot chocolates would she expect to sell?
- 11 A bag of oranges weighing  $\frac{8}{5}$  kg was divided among 6 children. What was the mass of the

oranges given to each child?



- 12 Tricia attends a 3-hour painting class. The instructor uses  $\frac{12}{14}$  of an hour to discuss colour mixing and then allows students to use the rest of class time to complete a painting. How much time does Tricia have left to finish her work?
- 13 Katrina and Luigi ordered a pizza to share. Katrina ate  $\frac{3}{7}$  of the pizza while Luigi ate  $\frac{1}{2}$ .
- a Who ate more pizza?
  - b What fraction of the pizza did they eat together?
  - c What fraction of pizza is left?
- 14 Luke has a dice tin that is  $\frac{1}{4}$  full. He puts 8 extra dice in and the tin is now  $\frac{1}{3}$  full.
- a What fraction of the tin do the 8 extra dice take up?
  - b How many dice can fit inside the tin?
- 15 Victoria borrows a 360-page novel from the library. She reads  $\frac{3}{4}$  of the novel.
- a How many pages did she read?
  - b How many more pages does she have to read to finish her novel?
- 16 Christa is creating a 3 minute slide show with pictures from her holiday. Each picture will be displayed for  $4\frac{1}{2}$  seconds.
- a How many seconds are in 3 minutes?
  - b How many pictures can she show in 3 minutes?
- 17 Danielle is studying for three tests. She studies science for  $3\frac{1}{4}$  hours, English for  $\frac{2}{5}$  of an hour and visual arts for  $\frac{1}{4}$  of an hour. How many hours does she spend studying?
- 18 Homer works as an accountant and is at the office for  $7\frac{1}{2}$  hours each day. He eats lunch with his colleagues for  $\frac{5}{6}$  hours each day. How much time does he spend working in one day?
- 19 In a school debate, each speaker was allowed  $1\frac{6}{7}$  minutes. How many speakers participated if the debate went for 26 minutes?

**20** Paul delivers online orders to customers for  hours each weekday.

- a** Over the five weekdays, how many hours does he work?
- b** On Saturdays he works half the time of a weekday. How many hours does he work on a Saturday?
- c** How many hours in total does he work in one week if he does not work on Sundays?

**21** Han spent  $\frac{1}{8}$  of his money on a printer and  $\frac{1}{5}$  of his money on a table.

- a** What fraction of his total money did Han spend?
- b** What fraction of Han's money does he still have?

**22** In a survey of 270 people,  $\frac{1}{10}$  said their favourite sport was soccer, and  $\frac{1}{9}$  said their favourite sport was tennis.

- a** Calculate the total fraction of people who said their favourite sport was soccer or tennis.
- b** Calculate the fraction of people who did not put soccer or tennis as their favourite sport.
- c** Calculate the number of people who did not put soccer or tennis as their favourite sport.

**23** In a class with 40 students,  $\frac{2}{5}$  of the class are boys.

- a** What fraction of the class are girls?
- b** How many girls are in the class?

**24** An engineer is estimating the length and width of a room using the length of his steps. The length of the room is  $\frac{4}{5}$  of the width. He determined that the width is about 27 steps and that one of his steps is  $\frac{2}{3}$  m long. Find the length of the room.

**25** Two cars leave home at the same time travelling south. Car A travels at the rate of  $83\frac{1}{4}$  km/h while Car B is at  $88\frac{3}{4}$  km/h. If both cars continue to travel at these speeds in the same direction, how far apart will the two cars be at the end of 4 hours?

**26** About  $\frac{3}{10}$  of the cars in a city are less than 5 years old. About  $\frac{1}{5}$  of the cars are between 5 and 10 years old. If there is a total of 1200 cars registered in the city, find the number of cars that are greater than 10 years old.

**27** The mall parking lot has 800 parking spaces.  $\frac{1}{4}$  of the parking spaces are for cars. On a particular day there were 200 cars and some trucks in the parking lot. The parking lot was  $\frac{5}{7}$

